Final deliverables: (1) a paper or cardboard prototype and (2) a simple proof of concept (PoC): a small demo of your core idea that shows how it looks and responds.

Week 1 (Oct 11-16): Understand your users (Needfinding)

- Contextual inquiry: watch or talk to real users in the setting where your product would be used (a desk, a kitchen, a hospital unit, etc.).
- Make an empathy map: what do users say, think, do, and feel?
- Write 1-2 short personas based on what you learned.
- *Output:* a one-page summary of who your users are and what problem you're solving.
- A short viva follows this stage and every group member should be able to explain the findings.

Week 2 (Oct 19-23): Build and test a paper prototype

1. Define goals: list the 3-5 things users most need to do, and the features and content needed to support them.
2. Sketch the big picture: map all your screens or steps on one page, then draw the detailed version of each one.
3. Show feedback: include what happens when the user waits, makes a mistake, or finishes a task. Keep labels and buttons consistent across screens.
4. Storyboard: a comic-style series of scenes showing someone using it from start to finish.
5. Test with at least 3 people: act as the "computer" yourself by swapping paper screens and responding to their taps. Note where they get confused, then change the prototype.

Weeks 3-4 (Oct 26 - Nov 6): Build the proof of concept

- Build a working version of your one or two most important features. Depth in a few features beats a shallow version of everything.
- Hardware projects (topics 1-2): assemble the sensors and microcontroller, and get one clear input to produce one clear response.
- Screen-based projects (topics 3-6): build a clickable prototype in Figma or similar, or a simple app.

Weeks 5-6 (Nov 7-16): Evaluate

1. *Usability testing:* give users real tasks and watch where they struggle.
2. *Cognitive walkthrough:* go through each task step by step, asking "Will the user know what to do here?"
3. *Heuristic evaluation:* check your design against standard usability rules. Swap with a peer group and evaluate each other's work.

Week 7 (Nov 17-20): Final showcase

- Bring your paper/cardboard prototype and your PoC.